



United International University (UIU)
Department of Computer Science and Engineering (CSE)
Course: VLSI Design (CSE 4327)

Experiment 1: Schematic Design of Standard Cells

Objectives

1. To understand the concept of a standard cell and the role of a schematic diagram in the IC design flow.
2. To become familiar with the Cadence Virtuoso Schematic Editor environment.
3. To create a new design library, build the transistor-level schematic of a CMOS inverter, and generate its schematic symbol.
4. To build a test bench for the inverter and verify its functionality using DC and transient analysis in Spectre (ADE L).

Introduction

A standard cell is a group of transistors and interconnect structures that implements a Boolean logic function, e.g., AND, OR, inverter, flip-flop, etc. The initial design of a standard cell is usually developed at the transistor level, in the form of a schematic diagram.

A schematic diagram shows the function of a circuit with little emphasis on its physical characteristics. Its primary goal is to display the functional relationships between components such as MOSFETs, so that the circuit can be easily analyzed. Schematics for electronic circuits are prepared by designers using EDA tools such as Schematic XL of Virtuoso from Cadence. These are usually integrated into the overall IC design flow and linked to other EDA tools (such as the Analog Design Environment, ADE) for verification and simulation of the circuit under design.

Note: In this experiment, you will design a CMOS inverter as a standard cell - starting from creating a design library, to drawing the transistor-level schematic, generating its symbol, and finally verifying its behaviour through simulation.

Part 1: Getting Started

1. Open a terminal and switch to the C-shell; this sets up your Cadence tools environment variables.
`csh`
2. Create a lab directory and a work sub-directory; also create simulation and states sub-directories under work:
`mkdir lab1`
`cd lab1`
`mkdir work`
`cd work`
`mkdir simulation states`

3. Copy the cds.lib file provided by your instructor into your work/ directory:
`cp <path-to-provided-file>/cds.lib_90 cds.lib`
4. We will use the virtuoso tool from Cadence to create our schematic. Start the tool:
 virtuoso &
5. This opens the Common Interpreter Window (CIW) and the Overview window. Read the Overview window to learn about the Virtuoso updates, then close it.



Figure 1: Virtuoso CIW (Common Interpreter Window).

Part 2: Creating a New Library

6. Click on Tools → Library Manager in the CIW. This opens the Library Manager window.

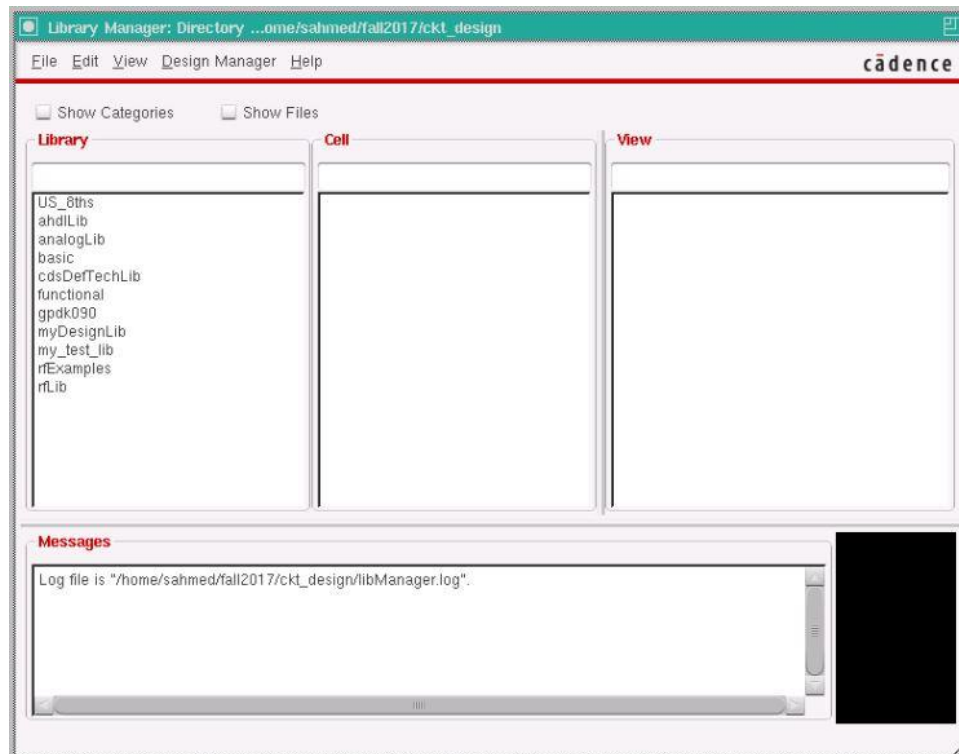


Figure 2: Library Manager window.

7. Click on File → New → Library in the Library Manager window. This opens the New Library form.
8. In the Name field of the New Library form, give a design library name (e.g., myDesignLib). Make sure the Directory field points to your work area. Click OK.



Figure 3: New Library form.

9. The “Technology File for New Library” form appears. Select the option “Attach to an existing technology library” and click OK.



Figure 4: Technology File for New Library form.

10. The “Attach Library to Technology Library” form appears. Select gpdk from the Technology Library field. Click OK.



Figure 5: Attach Library to Technology Library form.

11. This creates the design library, which now appears in the Library Manager under 'Library'.
12. Right-click on the newly created design library and select 'Properties'. This opens the Library Property Editor. Make sure gpdK is listed as the techLibName.

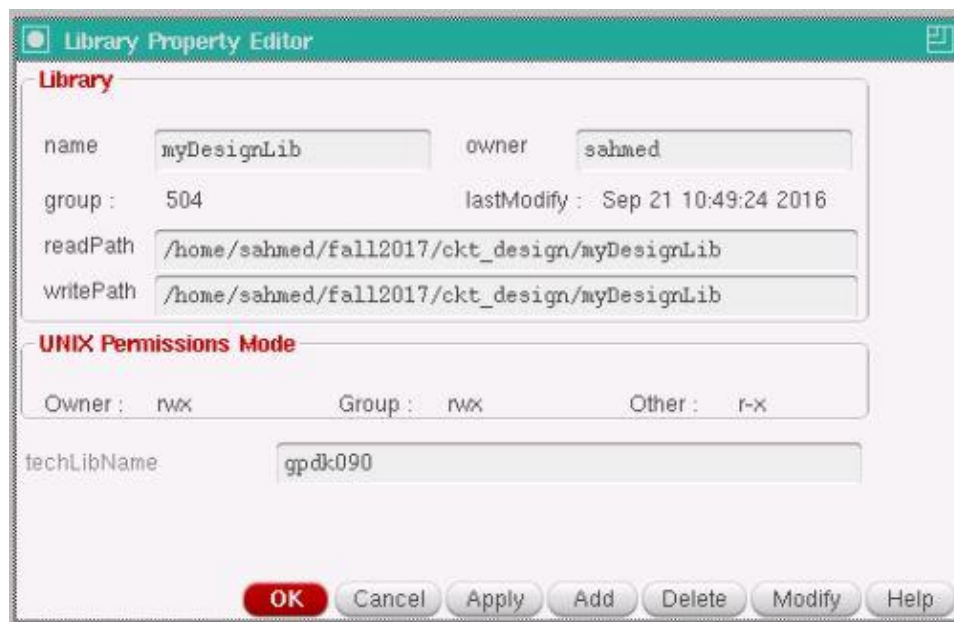


Figure 6: Library Property Editor.

Part 3: Creating a Schematic Cellview

13. Click on File → New → Cellview in the Library Manager window. The New File form opens.
14. Make sure Library is set to your design library, name the Cell as inverter, and set View and Type as schematic. Choose 'Schematic XL' for the Open with field and check 'Always use this application ...'. Make sure the Library path file is set to the cds.lib in your work area. Click OK.



Figure 7: New File form for creating the inverter schematic cellview.

15. A blank schematic window for the inverter design appears. Make sure the library name matches the one you created earlier.

Part 4: Adding Components to the Schematic

16. Click on Create → Instance (or press i). The Add Instance window appears.

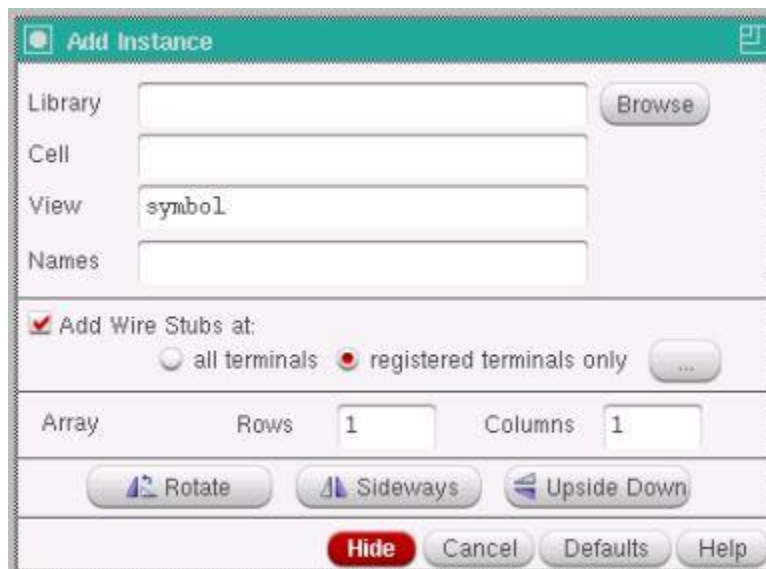


Figure 8: Add Instance window.

17. Click Browse next to Library. The Library Browser window appears. Select gpdk under Library, nmos1v under Cell, and symbol under View.

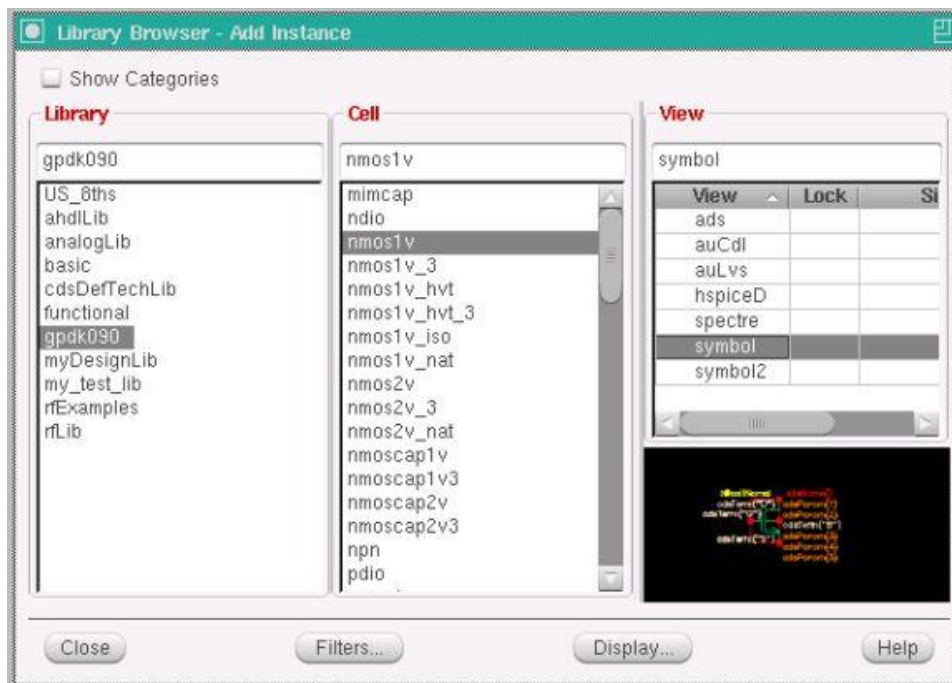


Figure 9: Library Browser - selecting the nmos1v symbol.

18. The Add Instance window updates. Set Total Width and Finger Width to 240n and leave Length at 100n. Review the other parameters in the Add Instance window to familiarize yourself with them.



Figure 10: Add Instance window with NMOS parameters set.

19. Move your cursor to the schematic window and left-click to place the component.

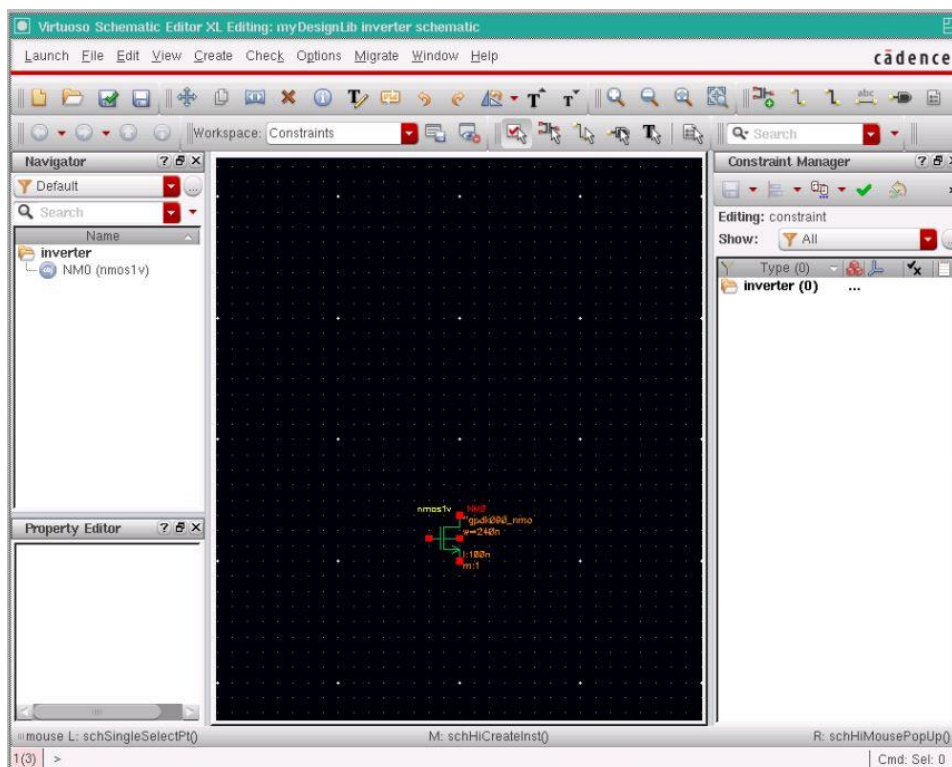


Figure 11: NMOS instance placed in the schematic window.

20. Repeat the process for the PMOS transistor: select gpdk under Library, pmos under Cell, and symbol under View. Set Total Width and Finger Width to 720n and Length to 100n. Move the cursor to the schematic window and left-click to place the component.

Library Name	Cell Name	Properties
gpdk	pmos	Total width = 720n
gpdk	nmos	Total width = 240n

21. You can change the parameter values of placed components using Edit → Properties → Objects. Use Edit → Move and Edit → Rotate to reposition/rotate components in the schematic window.

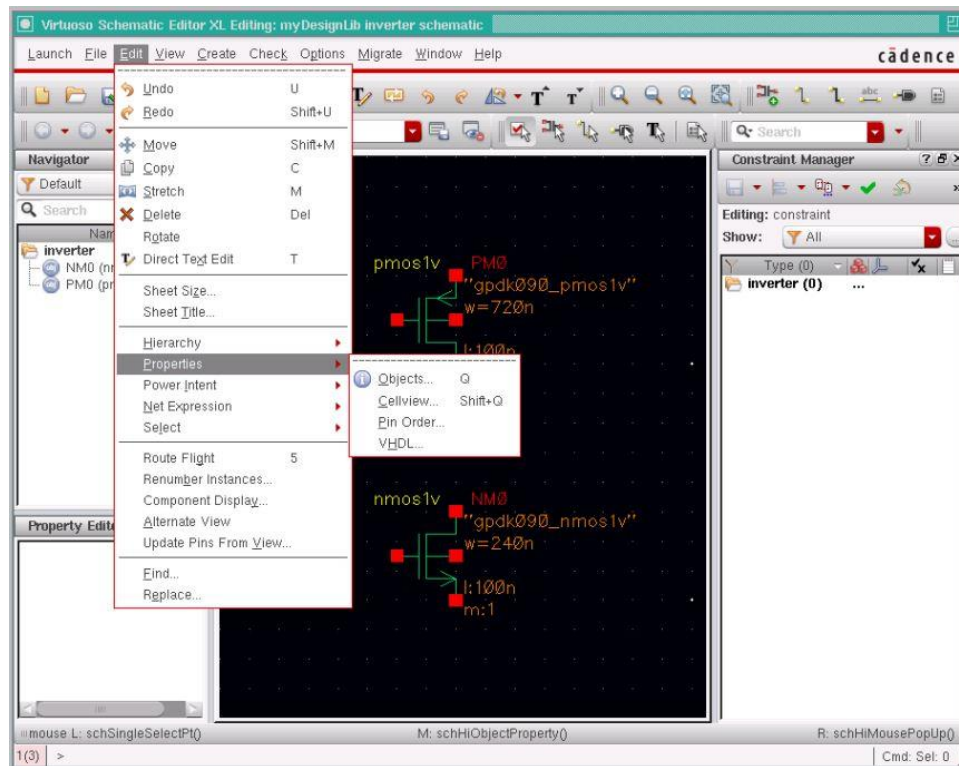


Figure 12: Edit → Properties menu, with PMOS and NMOS instances placed.

Part 5: Adding Pins to the Schematic

22. Click on Create → Pin (or press p). The Add Pin window appears.
23. Enter “in Vdd Vss” as Pin Names and select Direction as Input. Move the cursor to the schematic window and left-click to place the pins one at a time. Move/Rotate them as before if needed.



Figure 13: Add Pin window - adding the input pins (in, Vdd, Vss).

24. Enter “out” as Pin Name and select Direction as Output. Move the cursor to the schematic window and left-click to place the pin. Move/Rotate as needed.

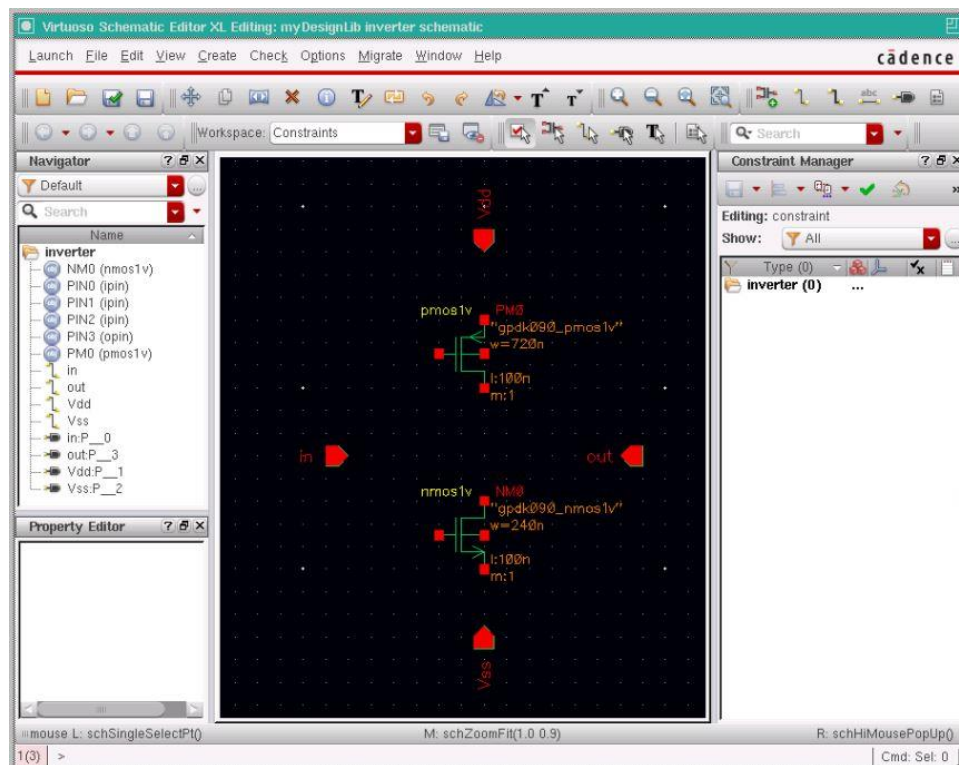


Figure 14: Completed pin placement - in, out, Vdd and Vss.

25. Select Cancel from the Add Pin window.

Part 6: Adding Wires to the Schematic

26. Click on Create → Wire (narrow) (or press w). Connect the different components and pins in the schematic window using the left mouse button. Press ESC after completing the connections.

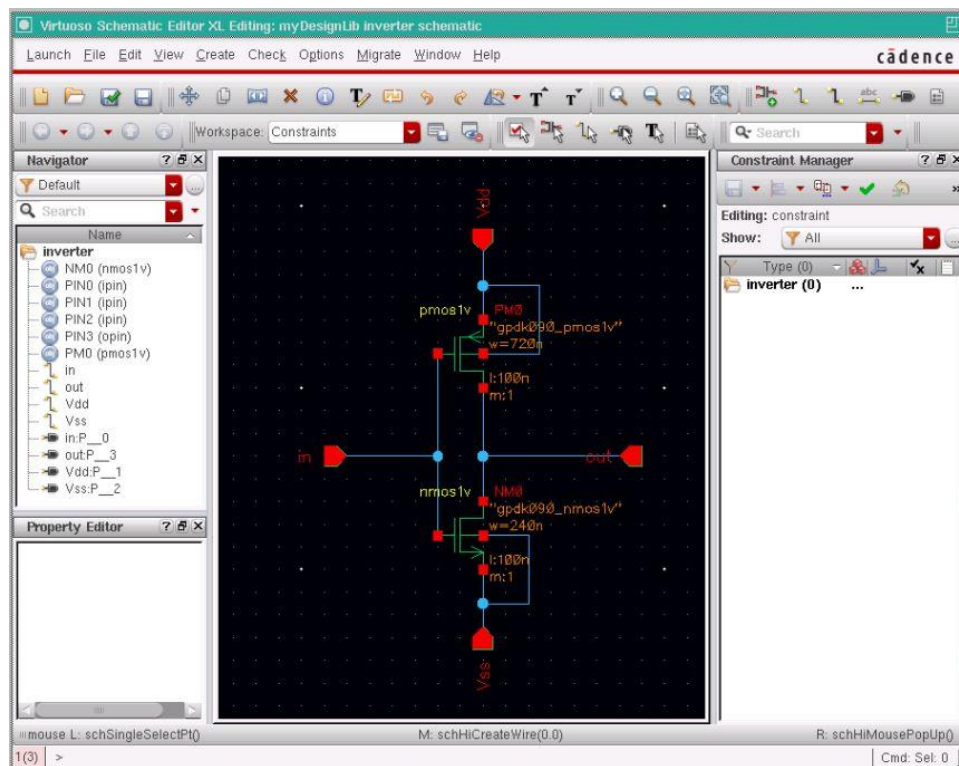


Figure 15: Wired inverter schematic.

Part 7: Saving the Design

27. Click the Check and Save icon in the Schematic window. It is very important to observe the CIW for any errors.

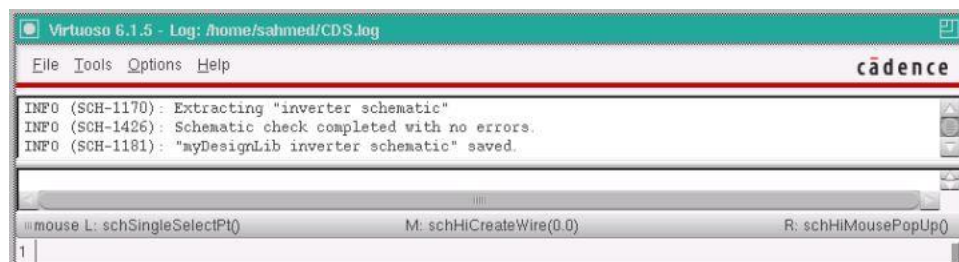
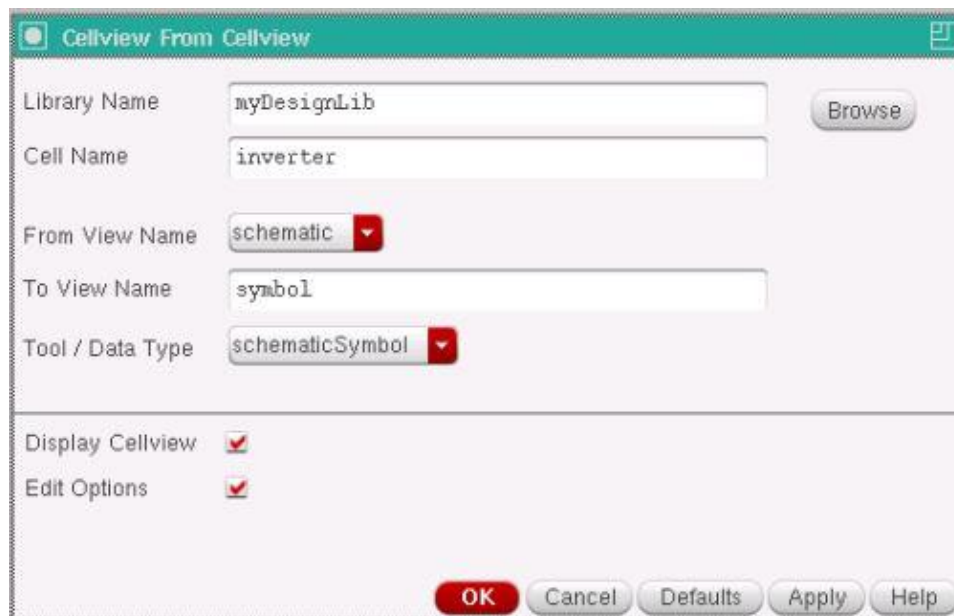


Figure 16: CIW confirming the schematic was checked and saved with no errors.

Part 8: Symbol Creation

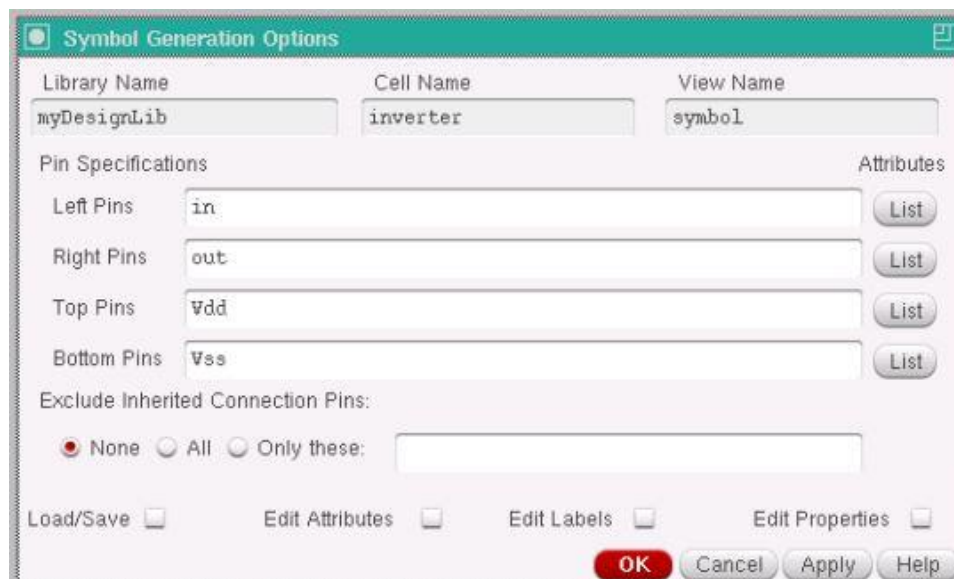
28. Click on Create → Cellview → From Cellview in the Schematic window. The “Cellview From Cellview” window appears.
29. Verify that From View Name is set to schematic, To View Name is set to symbol, and Tool/Data Type is set to schematicSymbol. Click OK.



The "Cellview From Cellview" dialog box is shown. It has a title bar with a green icon and the text "Cellview From Cellview". The main area contains several input fields and checkboxes. The "Library Name" field is set to "myDesignLib" with a "Browse" button to its right. The "Cell Name" field is set to "inverter". The "From View Name" field is set to "schematic" with a dropdown arrow. The "To View Name" field is set to "symbol". The "Tool / Data Type" field is set to "schematicSymbol" with a dropdown arrow. Below these fields are two checkboxes: "Display Cellview" and "Edit Options", both of which are checked. At the bottom of the dialog are five buttons: "OK", "Cancel", "Defaults", "Apply", and "Help".

Figure 17: Cellview From Cellview form.

30. The “Symbol Generation Options” window appears. Arrange the pins the way the symbol should have them, i.e., in as Left Pins, out as Right Pins, Vdd as Top Pins, and Vss as Bottom Pins. Click OK.



The "Symbol Generation Options" dialog box is shown. It has a title bar with a green icon and the text "Symbol Generation Options". The main area contains several input fields and checkboxes. The "Library Name" field is set to "myDesignLib", the "Cell Name" field is set to "inverter", and the "View Name" field is set to "symbol". Below these fields is a section titled "Pin Specifications" with four rows: "Left Pins" set to "in", "Right Pins" set to "out", "Top Pins" set to "Vdd", and "Bottom Pins" set to "Vss". To the right of each row is a "List" button. Below the "Pin Specifications" section is a section titled "Exclude Inherited Connection Pins:" with three radio buttons: "None" (selected), "All", and "Only these:". To the right of the "Only these:" radio button is an empty text field. At the bottom of the dialog are four checkboxes: "Load/Save", "Edit Attributes", "Edit Labels", and "Edit Properties", all of which are unchecked. At the bottom right of the dialog are five buttons: "OK", "Cancel", "Apply", and "Help".

Figure 18: Symbol Generation Options form.

Part 9: Editing the Symbol

31. A Symbol Generation window appears containing a default shape for the inverter.

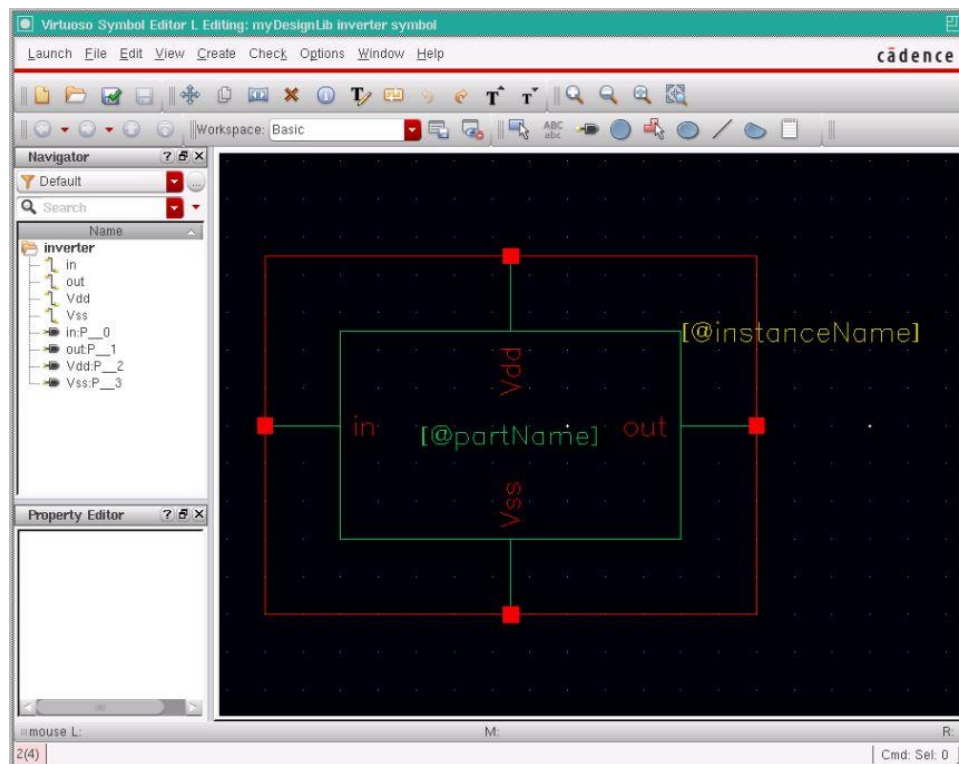


Figure 19: Default generated symbol shape for the inverter.

32. Left-click on the green rectangle to select it and press Delete to remove it. Similarly, remove the outer red rectangle.
33. Click Create → Shape → Line (or click the line icon) and draw a triangle shape. Press ESC.
34. Click Create → Shape → Circle (or click the circle icon) and draw a circle at the tip of the triangle shape (representing the inverter's bubble). Press ESC.

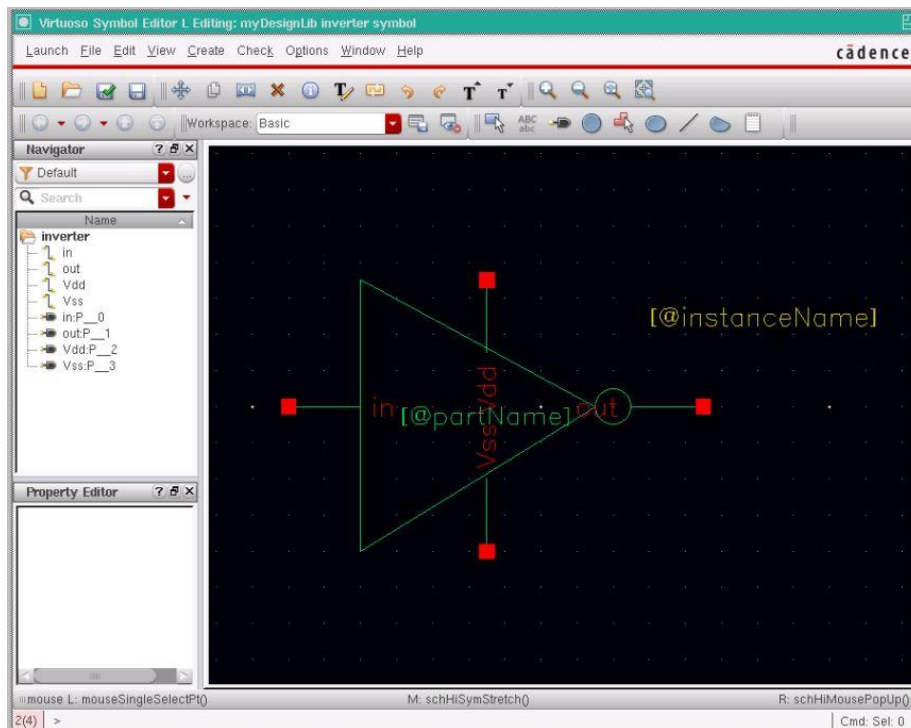


Figure 20: Inverter symbol redrawn as a triangle with an output bubble.

35. Click Create → Selection Box; a Selection Box form appears. Click Automatic. A new red selection box is added automatically.

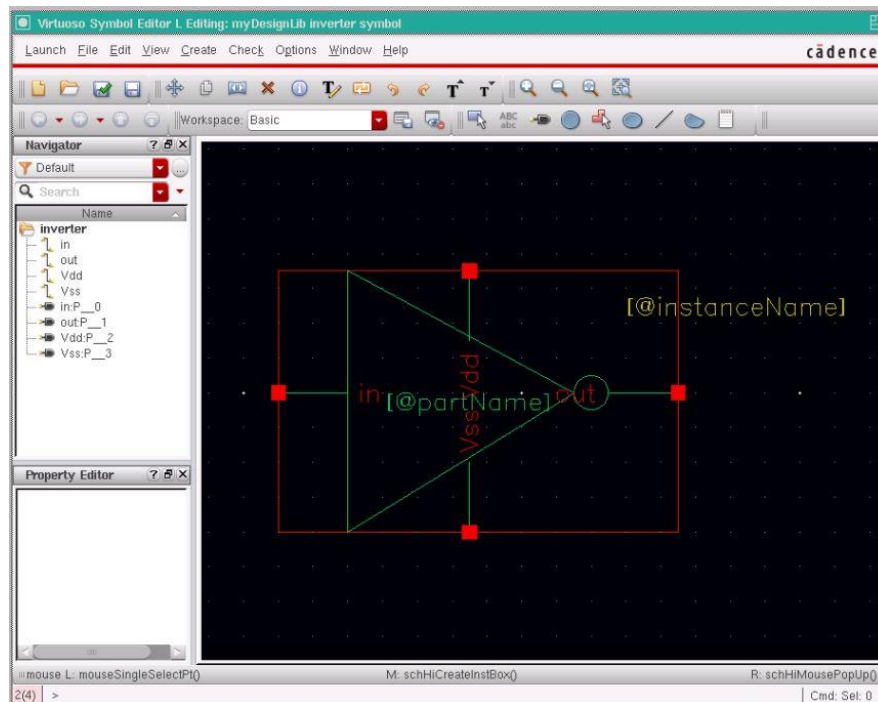


Figure 21: Automatic selection box added around the symbol.

36. Click the Save icon to save the symbol, then click File → Close to close the Symbol view window.

Part 10: Building a Test Design

37. Click on File → New → Cellview in the CIW or Library Manager.
38. In the New File form, set Library to your design library, Cell to inverter_test, View to schematic, Type to schematic, and Open with to Schematics XL, and select 'Always use this ...'. Make sure the Library path file is set to cds.lib in your work area. Click OK.



Figure 22: New File form for the inverter_test schematic.

39. A blank Schematic Editor window for the inverter_test design appears.

Building the Test Circuit

40. Follow steps 16 through 19 above to build the inverter_test circuit, adding the following components:

Library Name	Cellview Name	Properties
<your design library>	inverter	Symbol
analogLib	vpulse (as input signal)	voltage1 = 0, voltage2 = 1.8, period = 20n, pulse width = 10n
analogLib	vdc	DC voltage = 1.8
analogLib	gnd	-

41. Wire your schematic to connect the source, the inverter, and the supplies.

42. Click Create → Wire Name (or press L) to name the input (Vin) and output (Vout) wires.

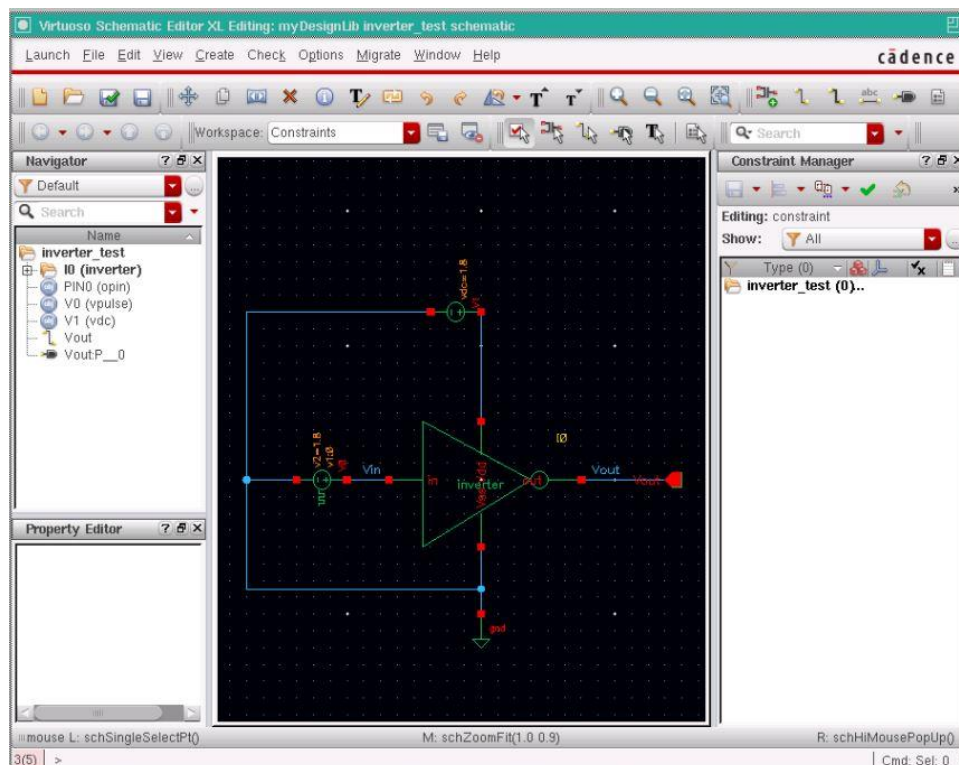


Figure 23: Completed inverter_test schematic - vpulse source, inverter symbol, vdc supply and ground, with Vin/Vout labelled.

43. Click the Check and Save icon to save the design.

Part 11: Analog Simulation with Spectre

44. Click on Launch → ADE L in the inverter_test Schematic window.

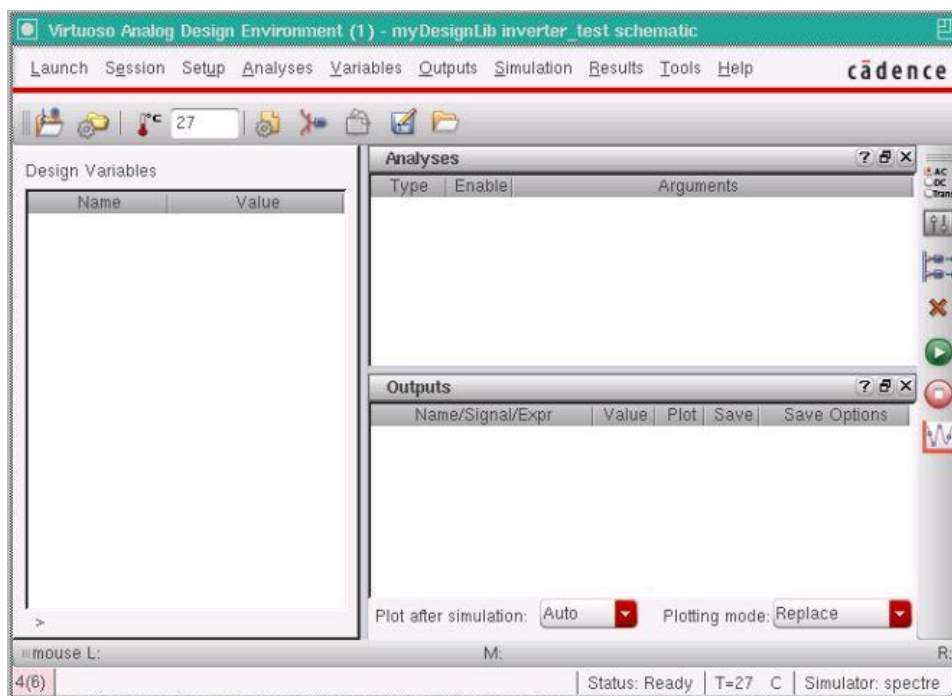


Figure 24: Virtuoso Analog Design Environment (ADE L) window.

45. The Virtuoso Analog Design Environment (ADE) simulation window appears. Click on Setup → Simulator/Directory/Host in the ADE window.
46. In the “Choosing Simulator” form, set the Simulator field to Spectre (not SpectreS) and Project Directory to your local simulation directory. Click OK.

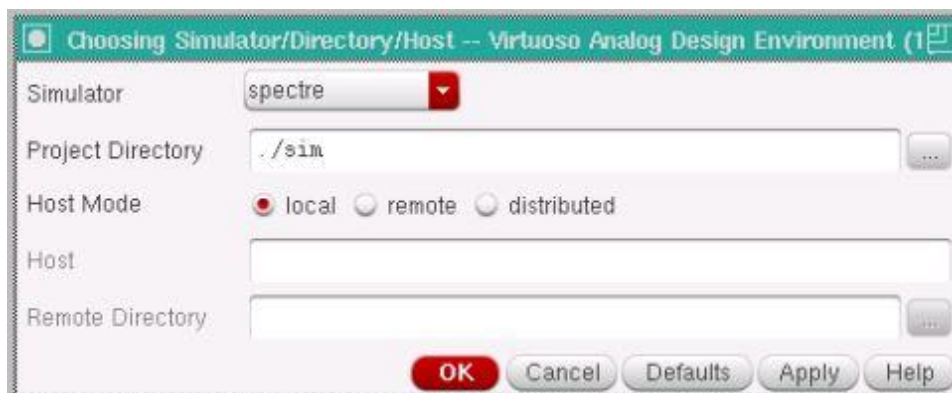


Figure 25: Choosing Simulator/Directory/Host form.

47. Click on Setup → Model Libraries in the ADE window.
48. In the “Model Library Setup” form, if not added by default, use the Browse button to add `.../gpd_k_dir/models/spectre/gpd_k.scs` and select the section type as NN next to the `gpd_k.scs` file. To view the model file, highlight the Model Files field and click the Edit File icon. Once satisfied, click OK.



Figure 26: Model Library Setup form.

Choosing Analysis

49. Click on Analyses → Choose in the ADE window. The “Choosing Analyses” form appears.

50. To set up the transient analysis:

- a. Select tran in the Analysis section.
- b. Set the Stop Time as 200n.
- c. Click on moderate and Enabled, then click Apply.

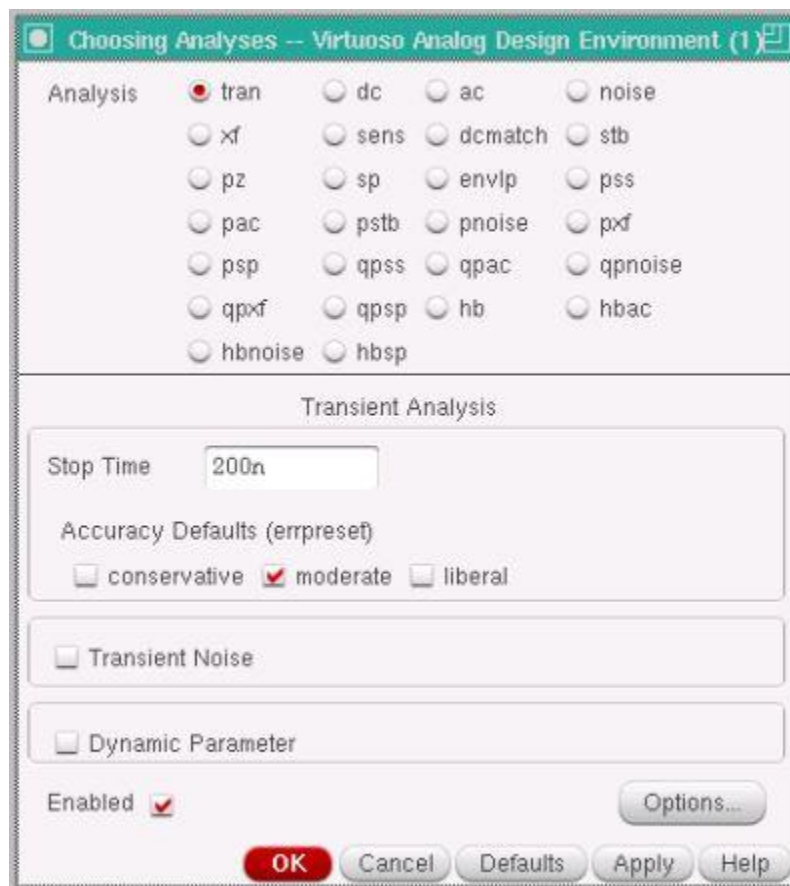


Figure 27: Transient analysis setup in the Choosing Analyses form.

51. To set up the DC analysis:

- a. Select dc in the Analysis section.

- b.** Turn on Save DC Operating Point in the DC Analysis section.
- c.** Turn on Component Parameter in the Sweep Variable section.
- d.** Click Select Component; this takes you to the test schematic window.
 - v. Select the vpulse source in the test schematic window.
 - vi. Select DC voltage in the “Select Component Parameter” window and click OK.



Figure 28: Select Component Parameter window - choosing the DC voltage parameter of the vpulse source.

- vii. Enter start voltage as 0 and stop voltage as 1.8.
- viii. Select Enabled, then click Apply.

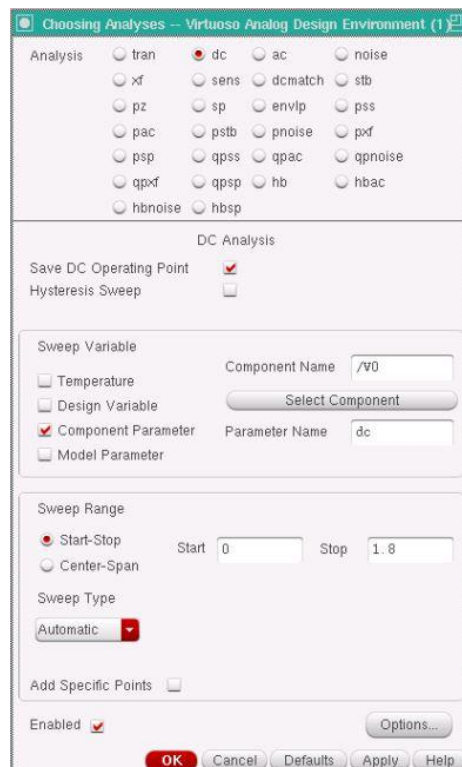


Figure 29: DC analysis configured as a sweep of the vpulse DC voltage from 0 to 1.8 V.

52. Click OK in the Choosing Analyses form.

Setting Design Variables

53. Click Variables → Edit Variables in the ADE window.

54. In the “Editing Design Variables” window, click Copy From under Cellview Variables. The design is scanned and any variables found are listed - for example, a wp variable representing the PMOS width. Highlight the variable, enter its value (e.g., 720n) in the Value(Expr) field, and click Change. Click OK to exit.

Note: If your design does not use any design variables (i.e., all widths were entered directly as instance parameters), this step is not required.

Selecting Signals for Plotting

55. Click on Outputs → To Be Plotted → Select on Schematic in the ADE window.

56. Click on the input net Vin and the output net Vout of the inverter_test schematic, then press Esc.

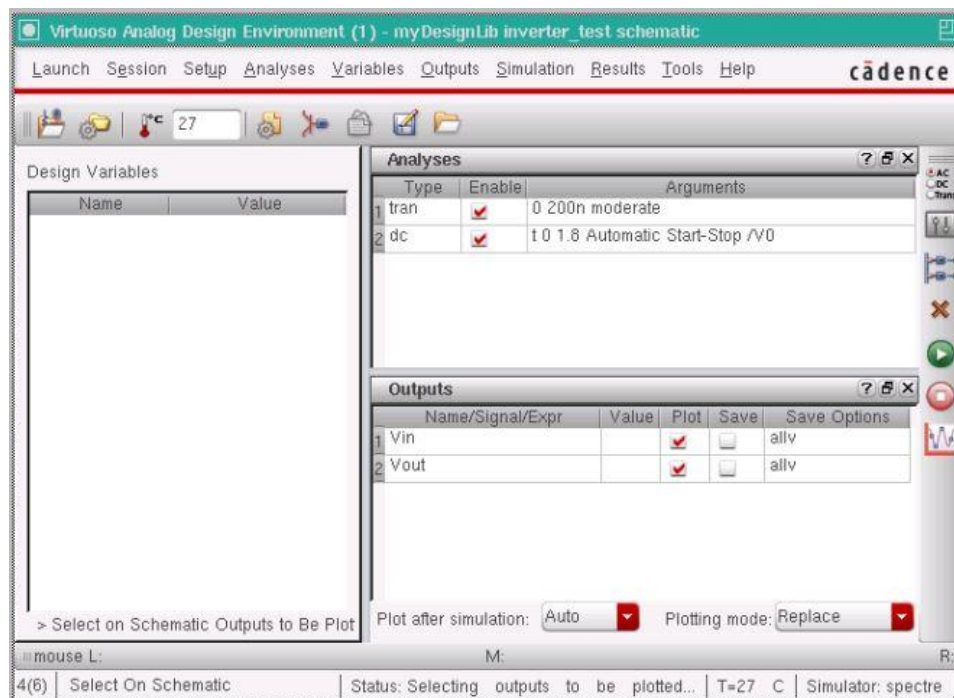


Figure 30: ADE window with the transient and DC analyses enabled and Vin/Vout selected as outputs to be plotted.

Running the Simulation

57. Click on Simulation → Netlist and Run in the ADE window to start the simulation.

58. When the simulation finishes, the transient and DC response plots appear along with the log file.

59. Click on the Split Current Strip icon to separate the waveform plots.

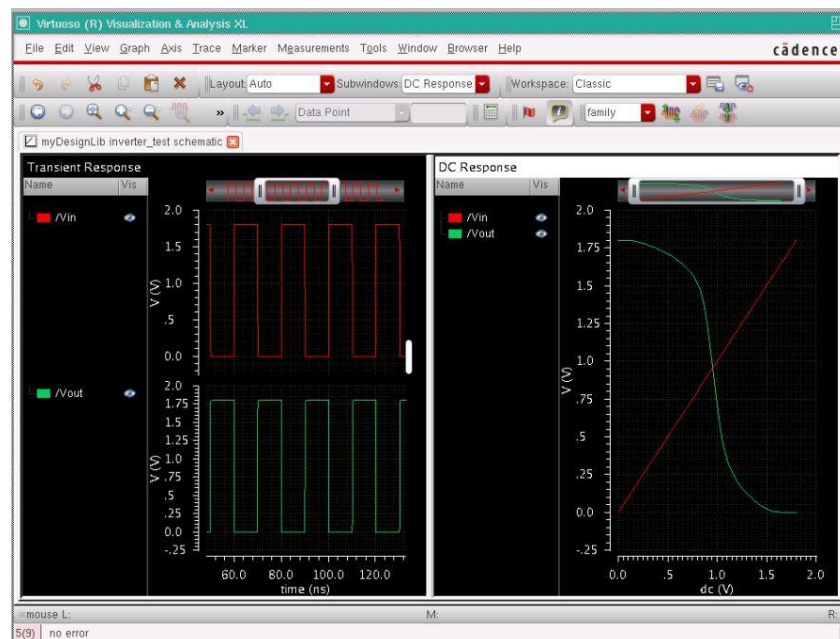


Figure 31: Transient response (V_{in} , V_{out}) and DC voltage-transfer response of the inverter.

Note: From the DC response (V_{out} vs. sweep of V_{in}), locate the switching threshold voltage V_M - the point where $V_{in} = V_{out}$ on the transfer curve - and record it in your report along with the rise/fall times observed in the transient response.

Saving the Simulator State

60. Click on Session → Save State in the ADE window.

61. In the “Saving State” form, set the Save State Directory field to your states directory (e.g., `.../lab1/work/states`), give a name to the Save As field (e.g., `state_inverter`), and make sure all options are selected under What to Save. Click OK to save the state.

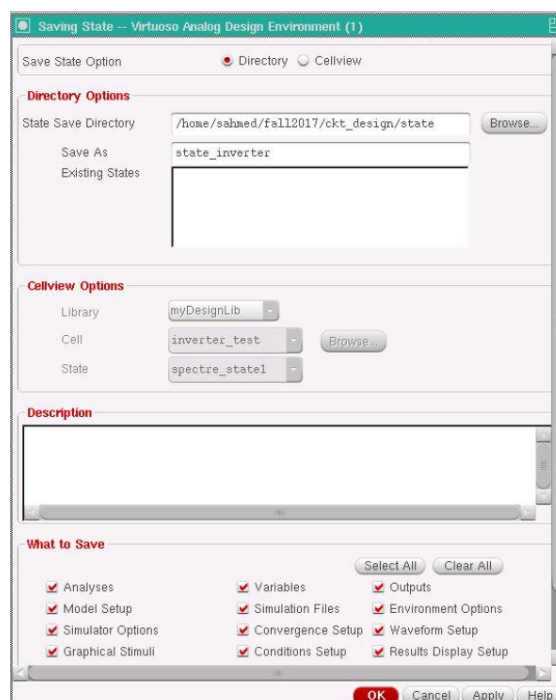


Figure 32: Saving State form.

Loading the Simulator State

62. Click on Session → Load State in the ADE window.
63. In the “Loading State” form, choose the correct Library, Cell, and Simulator (i.e., your design library, inverter_test and spectre respectively). Set the State Load Directory and State Name to the correct directory and name. Click OK to load the state.

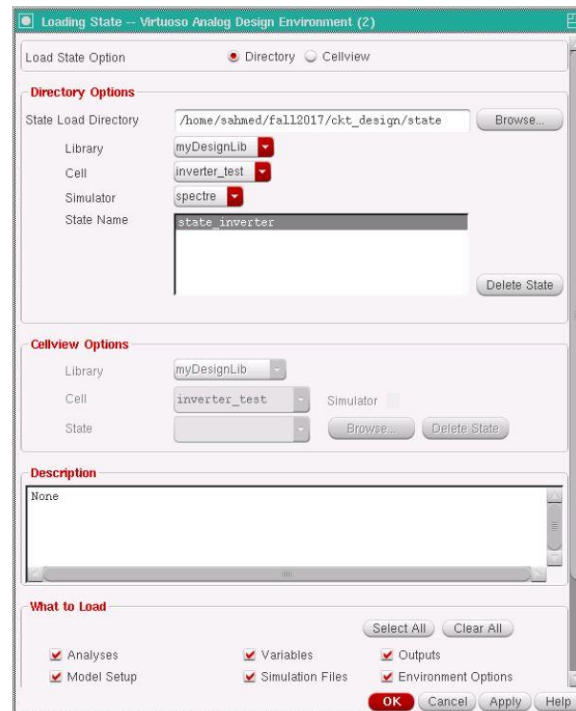


Figure 33: Loading State form.

Lab Report / Submission Checklist

1. A screenshot of your completed inverter schematic (with pins and wires labelled).
2. A screenshot of your generated inverter symbol.
3. A screenshot of the inverter_test schematic used for simulation.
4. The transient response plot (Vin and Vout vs. time) with the pulse period and pulse width clearly visible.
5. The DC voltage-transfer characteristic (VTC) plot, with the switching threshold voltage VM marked.
6. A short discussion (150-200 words) explaining how the PMOS-to-NMOS width ratio (720n : 240n) relates to balancing the pull-up and pull-down strengths of the inverter.

Credits

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